

Predictive-Maintenance Air Purifier

Innovation dossier

1. What is it?

A concept for the real 'Reliability / stopped working' friction, built with the PREDICT design move - reactive -> anticipatory.

2. Physical architecture

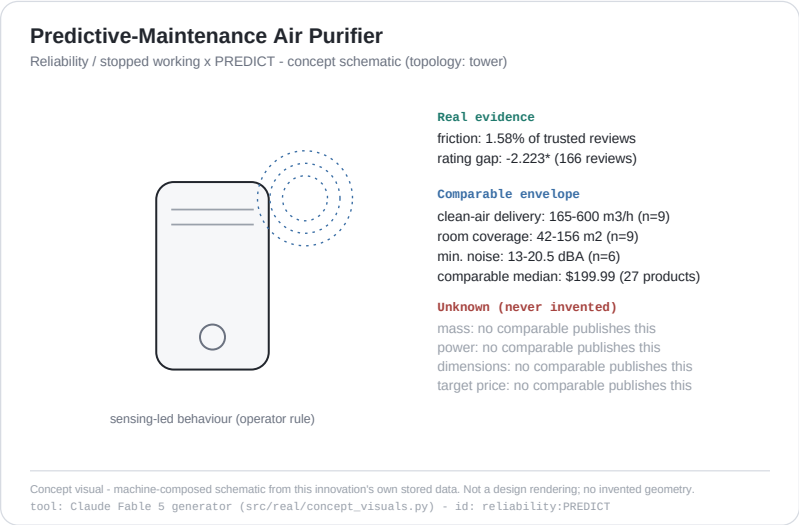


Figure: [Concept visual](#)

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

artifact visual-reliability:PREDICT - /concept-visuals/reliability_PREDICT.svg

Product archetype

- room scale:** Unknown - no real data in this pipeline for this field.
- mobility:** Unknown - no real data in this pipeline for this field.
- size class:** Unknown - no real data in this pipeline for this field.
- power class:** Unknown - no real data in this pipeline for this field.
- noise ambition:** Unknown - no real data in this pipeline for this field.
- maintenance burden:** Unknown - no real data in this pipeline for this field.
- filter service dependency:** Unknown - no real data in this pipeline for this field.
- connected:** Likely - operator PREDICT (reactive -> anticipatory) requires sensing/data.
- form factor:** tower

Design rule - each populated field follows from the operator's own fixed definition, never from evidence

System architecture

Unknown - no system-architecture dataset exists; the archetype above is a design rule from the operator's own definition.

Method choice

PREDICT - Reactive -> anticipatory.

Method choice - authored design vocabulary

3. Why it exists

Reality Observed

1.58% of trusted reviews (166 reviews across 60 real products) carry this friction, with a -2.223* average rating gap. Signal: 'Stopped working' or failing early is the most severe complaint theme by average rating gap.

Households whose real reviews carry this friction: 166 reviews across 60 real products (85.5% verified purchase, 2006-04-20-2023-08-27).

Unknown - no persona/behavioural dataset exists in this pipeline; only the review evidence above is real.

Transformation Method choice

METHOD CHOICE, not evidence: the authored operator table assigns PREDICT - "Reactive -> anticipatory." - to this friction. Analyst design judgment, applied deterministically.

Product consequence Feasibility precedent

Feasibility precedent (from the trend corpus, not a consumer-evidence consequence): Two real, already-standardized frameworks exist to build on directly: TC-R02 (ENERGY STAR already requires measured, verified ongoing performance criteria - a real regulatory template for an ongoing-performance-verified claim) and TC-R08 (Matter's Air Quality Sensor cluster is an already-standardized connectivity layer that could carry a self-diagnostic/runtime health signal without inventing a new protocol).

4. Evidence and lineage

Parent paths

No parent paths - this innovation cites no tension or assumption paths; its grounding is the friction evidence in section 3, why it exists.

evidence ids: taxonomy:reliability, TC-R02, TC-R08

category assumptions: No category assumption shares a paper with this theme's signal evidence.

Design DNA

F consumer friction (taxonomy:reliability) Present

Real Amazon review-text friction theme (see Consumer Pain methodology).

O design operator Method choice

Authored operator vocabulary applied deterministically (METHOD_CHOICE, not evidence) - see OPERATORS/THEME_OPERATORS in src/real/magic_box_real.py.

S signal (reliability) Present

'Stopped working' or failing early is the most severe complaint theme by average rating gap.

T scientific tension Missing / unverified

No real research tension shares a paper with this theme's signal evidence.

R rival gap Present

7 named real competitors measurably weaker on this theme. [Envion, EdenPURE, Himalayan, Egofine, Belita Amy, MOONQUAKE, ZOCALOK]

C versuni capability Missing / unverified

No real Versuni internal capability/org-readiness dataset exists in this pipeline.

A category assumption Missing / unverified

No category assumption shares a paper with this theme's signal evidence.

E economic condition Present

Real price-weighted exposure computed for this theme (\$26,357.86).

5. Comparable products

comparable set: 10 verified official Versuni/Philips air-purifier families (room-scale)
clean-air delivery rate (CADR): 165-600 m3/h across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)
room coverage: 42-156 m2 across 9 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)
minimum noise: 13-20.5 dBA across 6 real comparables
source: individually verified official Versuni/Philips product pages (data/visual/product_images.json)

6. Engineering envelope

metric	value from verified comparables	epistemic type
clean-air delivery rate (CADR)	165-600 m3/h (n=9 comparables)	Observed comparable
room coverage	42-156 m2 (n=9 comparables)	Observed comparable
minimum noise	13-20.5 dBA (n=6 comparables)	Observed comparable
target mass	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target power	No verified comparable publishes this value - left unknown, never estimated.	Unknown
target dimensions	No verified comparable publishes this value - left unknown, never estimated.	Unknown

7. Economics

price-weighted exposure: \$26357.86

SUM of real observed listed prices across affected reviews with a known price. This is a RELATIVE exposure indicator, not a revenue or market-size estimate - it has no units-sold, conversion-rate, or time-period basis. It answers 'which friction touches more expensive products' not 'how much money is at stake per year.'

comparable market median: \$199.99 (n=27 products) - not this concept's price

MEDIAN real listed price across the 27 distinct real products affected by this friction that have a known price. This is what products in this segment actually cost today - not a proposed price for a new concept, which this evidence cannot establish.

target price: Not modelled - No concept target-price model exists - only the comparable market median (\$199.99, 27 real products) as reference. A target price would be invented.

8. Prototype

State: Concept visual

Concept visual means a schematic exists, nothing more: no digital model, no physical prototype, no test data.

Machine-composed schematic from the innovation's stored fields only (operator rule, friction stats, comparable envelope). Not a design rendering; a Fable render or drawing would also be CONCEPT_VISUAL, never a digital model or physical prototype.

9. Experiment and next learning

Next experiment

A real internal engineering estimate showing the self-diagnostic sensor suite costs materially more than a typical product refresh would lower this rating.

Kill criterion

No authored kill criterion yet for this innovation - only the formal case (see formal_case.abandon_signal) has a human-authored one; this innovation earns its own by developing further evidence or by a human decision to invest.

10. What could kill it?

Where this idea stands: Challenged

Critic verdict CHALLENGE - a real dimension pushes back.

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Uncertainties

- No real Versuni R&D-capacity or hardware-warranty-claims data - the rating reflects external precedent, not an internal build-cost estimate.

Critic dimensions

Evidence	Challenge	Gate passed but prevalence (1.58%) is only marginally above the materiality floor - thin evidence base.
Economic	Survive	Real price-weighted exposure (\$26,358) is at or above the corpus median (\$16,113).
Competitive	Survive	Real white space: 7 named real rivals are measurably weaker on this theme.
Human	Needs evidence	No real usability/human-factors evidence exists in this pipeline for this concept.
Physical	Needs evidence	No real lab/engineering feasibility data exists in this pipeline for this concept.
Versuni fit	Needs evidence	No real organizational-capability assessment exists in this pipeline for this concept.
Timing	Needs evidence	No real market-timing/window evidence exists in this pipeline for this concept.
Robustness	Needs evidence	No real sensitivity/robustness study exists in this pipeline for this concept.

critic overall: Challenge

Criteria results

Needs evidence: 36 - Pass: 11 - N/A: 4 - Challenge: 1

E3 Challenge Signal state SINGLE_SOURCE_FAMILY - real, but not yet independently corroborated.

Lineage and provenance

generated by: src/real/innovations_real.py
magic run input sha256: 0b3f9862e0e270a295eacd709182802e981090e817f4f1072000e37aa83f150c
funnel snapshot: d5d03b0d3856721f04d7d1c4ee7f2fe2a7d1df64ac4d2d4130bede07d2b13f61
figure source: /concept-visuals/reliability_PREDICT.svg (generated 2026-08-31T18:00:05.646592+00:00 by Claude Fable 5 generator - src/real/concept_visuals.py)
data source: data/processed/innovations_real.json + data/processed/funnel_real.json (deterministic; no invented fields)
dossier generator: src/real/generate_innovation_dossiers.py